

NDI — MOLECULAR NANOTECHNOLOGY: AN OPEN CAPABILITY FRAMEWORK

Extending the Definitions of Molecular Nanotechnology

A critique of the 1995 Feynman Grand Prize, and a generalized, characterization-first framework for productive nanosystems

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Status: this document does *not* propose to alter or replace the Foresight Institute Feynman Grand Prize. It extends the working *definitions* of molecular nanotechnology to accommodate approaches developed since 1995, and proposes a common characterization standard for judging them.

In one paragraph. The Feynman Grand Prize is the right north star with a 1995 specification. It measures the *motions* of a builder and the *logic* of a computer, silently presuming a stiff diamondoid machine, and it predates three decades of routes to atomically precise matter — scanning-probe mechanosynthesis, single-site catalysis, designed DNA and protein scaffolds, and templated crystalline growth. We first set out a one-page critique of the original specification, then extend the definition of molecular nanotechnology from an implementation to a **set of measurable capabilities** — positional precision, programmability, atomic precision, delivered force, elemental generality, product stiffness, and verifiability — that any of these approaches can be scored against. The emphasis throughout is characterization: how each capability is independently measured, to what uncertainty, and with what controls against spoofing.

Contents

1. A critique of the 1995 Feynman Grand Prize

The prize is the right target, specified for the wrong era.

The Feynman Grand Prize — \$250,000, established 1996, unclaimed — asks for two devices: a nanoscale robotic arm that positions to 0.1 nm and performs $\geq 1,000$ motions/s inside a 100 nm cube, and a nanoscale 8-bit adder inside a 50 nm cube, each delivered in 32 copies for destructive testing.^{1 2} It remains the field's canonical milestone and a faithful statement of intent: build matter to atomic specification under program control, and compute in the same volume. The criticism that follows is not of the ambition but of the *specification*, read as the rule-set for an objective, present-day competition.

It specifies an implementation, not a capability

The prize measures motions, positions, and rates — properties of one particular machine — rather than the products and forces that create value. Those numbers (0.1 nm, 1 kHz, raised “bumps” on an atomically precise surface) are trivial for a stiff covalent diamondoid device and effectively impossible for a soft one, so the specification covertly mandates a single chemistry. A capability the field actually wants — “place atoms of a specified element where directed, verifiably” — is nowhere stated as such.

It omits force — the quantity that does chemistry

Holding 0.1 nm against thermal noise requires a stiffness of only $k \approx k_B T / \langle x^2 \rangle \approx 0.41$ N/m, whereas exerting the ≈ 6 nN needed to drive a Ge–C mechanosynthetic reaction within that same 0.1 nm demands $k \approx 60$ N/m — a $\sim 145\times$ gap the prize never names.³ A device built exactly to the literal spec would deflect ~ 15 nm under a real reaction load ($150\times$ its own accuracy budget) and could not perform controlled mechanosynthesis. Position can be feedback-stabilized on a compliant structure; static reaction force cannot. The 1995 authors, inside the diamondoid paradigm, inherited stiffness and force “for free” from the material and never wrote them down.

It omits energy — the deciding figure of merit for the computer

The adder clause constrains logic and geometry but sets no energy-per-operation ceiling, which is the single quantity that decides whether molecular computing is ever useful. “A molecular adder” and “a molecular adder worth building” differ by exactly that omission.

It is procedurally underspecified, and binary

As a competition rule-set it leaves the decisive operational terms undefined: no measurement methods or metrology protocol, no named judging body, no operating environment (vacuum, cryogenic, ambient, liquid), no input-signal classes beyond a brief list, and no time-to-claim. It is also all-or-nothing: there is no credit for the intermediate productive-nanosystem capability that represents most of the real technical progress between here and the top of the mountain.

It is frozen in the 1995 solution space

Most consequentially for this document: the specification predates the methods that now plausibly reach atomic precision. Scanning-probe hydrogen-depassivation lithography and inverted-mode STM mechanosynthesis,^{4 5} positionally-controlled single-site catalysis, generative de novo protein and enzyme design at ångström precision,⁶ structural DNA nanotechnology with sub-nanometer addressability,⁷ and templated crystalline growth (MOFs/COFs, area-selective ALD) are all post-1995 and all relevant. A specification that recognizes only one of them cannot serve as the field's definition of success.

Conclusion. These are not defects to be patched into Foresight's prize, which retains real value as the stiff-covalent apex. They are the reason to extend the *definitions* of molecular nanotechnology — from

one machine's motions to a measurable set of capabilities, with a characterization standard rigorous enough to compare approaches that did not exist when the prize was written. That is the subject of the rest of this document.

2. Molecular nanotechnology as a set of capabilities

We replace “a 100 nm arm that positions to 0.1 nm” — a means — with a vector of capabilities that together define a **productive nanosystem**: a system that places and bonds matter at specified locations, under program control, to a stated precision, with the force its chemistry requires, across a stated range of elements, yielding a product of stated stiffness, all *independently verifiable*. Each axis is continuous and separately measurable, which is what makes partial credit and cross-approach comparison possible.

2.1 The capability axes

Axis	Definition	Why it matters / how it is bounded
Positional precision	RMS placement error of a feature relative to its target lattice or coordinate frame.	The headline 1995 number. Meaningful only when paired with the force present during placement.
Programmability	Ability to build a judge-specified, test-time-revealed target, not a single hard-wired product.	Distinguishes a general engine from a one-off artifact; the original arm was implicitly a programmable positioner.
Atomic precision	Defect rate — wrong-site, wrong-species, or missing placements per total placements.	“Atomically precise” must be a number (e.g. ≤ 1 defect per 10^4 – 10^6), not an adjective.
Delivered force	Static force the tool can source at the reaction site while holding selectivity under load.	The prize's missing quantity. Sets which chemistries are reachable; cannot be feedback-faked.
Elemental generality	Breadth of the periodic table the method can place and bond with control.	A single-element demonstration is a point; generality is the slope toward a manufacturing capability.
Product stiffness	Elastic modulus / class of material produced (soft organic $\rightarrow$ stiff inorganic, $\sim 10^2$ – 10^3 GPa).	Separates scaffolds that template matter from engines that build load-bearing, covalent structure.
Throughput & duty	Sustained positionally-controlled reactions per second, and for how long.	Converts a single placement into a manufacturing rate; the prize's 1,000 motions/s lives here.
Verifiability	Whether an independent party can confirm every claim above without trusting the entrant.	The axis the 1995 spec omitted entirely; the subject of Section 3.

Table 1. The capability axes that define a productive nanosystem. The 1995 prize fixed two of these (positional precision, throughput) and one output format, and left the rest implicit or unstated.

2.2 Four families of approach, on a common frame

Four broad families have matured since 1995. They are usually presented as rivals; on the capability frame they are **complementary** — each dominates a different region of the axes, and a credible APM program borrows from several. For each we state what it is, its current demonstrated state, and where it sits.

A. Scanning-probe / positional mechanosynthesis

A sharp tip (STM/AFM) removes, adds, or rearranges individual atoms on a surface. Hydrogen-depassivation lithography selectively desorbs H from Si(100):H to expose dangling bonds with single-atom abruptness, into which dopants (e.g. phosphine) are placed — the basis of atomic-precision advanced manufacturing now being integrated into conventional silicon middle-of-line fabrication.⁴ Inverted-mode STM has demonstrated spatially and chemically controlled C₂ donation to pre-patterned reactive sites, including stepwise polyynes assembly through successive C–C bond formation — mechanosynthesis in the literal sense.⁵ Non-contact AFM resolves individual covalent bonds, giving the field its imaging ground truth.⁸

- **Strengths:** highest positional precision and the only family that natively delivers covalent force; direct, single-atom verifiability by the same probe that builds.
- **Limits:** serial and slow (throughput axis); typically ultra-high-vacuum / cryogenic; elemental generality still narrow (Si, C, a handful of dopants).

B. Positionally-controlled single-site catalysis

Rather than forcing a reaction mechanically through its barrier, a single-site transition-metal catalyst lowers the barrier and is positioned to act at a chosen surface site. This is the proposed generality unlock: it replaces the force-heavy Ge→Si→C path with chemistry that needs far less delivered force (target ~100 pN–nN versus ~6 nN uncatalyzed) and, because catalysis spans much of the periodic table, broadens the reachable element set. *This is the central technical bet of the program and is presently a hypothesis-to-be-demonstrated* rather than a measured result; the force-reduction figure in particular must be established experimentally, not assumed.

- **Strengths:** the only family with a credible path to broad elemental generality at manufacturing-relevant force; compatible with the positional hardware of family A.
- **Limits:** least experimentally proven of the four for positional control; turnover, selectivity-under-load, and catalyst regeneration are open figures of merit.

C. Structural DNA and designed-protein scaffolds

Self-assembled DNA origami places guest features with sub-nanometer addressability and accuracy by modifying single staple strands,⁷ and generative protein design has crossed into atomic accuracy: RFdiffusion2 scaffolds all 41 sites of an enzyme-active-site benchmark (versus 16/41 for prior methods) and yields active de novo enzymes from reaction mechanisms, with crystal structures confirming ångström-level active-site precision.⁶ These are programmable positioning systems built from soft matter.

- **Strengths:** massively parallel (throughput) and highly programmable; ångström-scale placement of functional groups; aqueous, ambient operation.
- **Limits:** soft products (low stiffness axis) and negligible delivered force; best understood as templating/positioning scaffolds that can hold catalysts or nucleate inorganic growth, not as covalent builders themselves.

D. Templated and crystal-engineered inorganic growth

Metal-organic and covalent-organic frameworks (MOFs/COFs) self-assemble into atomically defined crystalline lattices with designed pore geometry; area-selective and atomic-layer deposition grow inorganic material with sub-nm thickness control and chemical site-selectivity; biotemplated routes use family-C scaffolds to nucleate inorganic phases. The product here is stiff and inorganic — the material class APM ultimately needs.

- **Strengths:** delivers stiff inorganic product at scale; crystallographically exact; broad and growing elemental palette.
- **Limits:** precision is statistical/lattice-enforced rather than arbitrarily addressable; weak on programmability (you get the lattice the chemistry wants, not an arbitrary specified object).

2.3 The capability scorecard

Placing the families on the axes makes the complementarity explicit and shows why no single approach “wins” — and why the 1995 specification, by recognizing only the diamondoid corner, scores most real progress as zero. Ratings are coarse (● strong, ◐ partial, ○ weak/absent) and indicate *current demonstrated* state, not ultimate potential.

Capability axis	Diamondoid (1995 ideal)	A. Scanning-probe	B. Single-site catalysis	C. DNA / protein	D. Templated inorganic
Positional precision	● (target)	●	◐	◐	○
Programmability	●	●	◐	●	○
Atomic precision	●	●	◐	◐	●
Delivered force	●	●	◐ (claimed)	○	○
Elemental generality	○	○	● (potential)	○	◐
Product stiffness	●	◐	◐	○	●
Throughput / duty	○	○	◐	●	●
Maturity (proven)	○ (none yet)	●	○	●	●

Table 2. Capability scorecard. The diamondoid ideal is strong everywhere but elemental generality and remains unrealized; the four post-1995 families each cover a different deficit. A productive-nanosystem program is read across the “delivered force × generality” corner only family B credibly addresses.

3. Characterization — the heart of the extension

Extending the definition is only credible if every capability claim is independently measurable. The original prize's deepest procedural gap was that it named no metrology; a material-agnostic, multi-approach definition makes metrology **harder, not easier**, because the same claim ("0.1 nm placement") must be adjudicated across vacuum-probe, aqueous-scaffold, and crystalline-growth entrants. This section is given in two layers: a strategic survey of the available techniques (3.1), and a per-capability acceptance protocol that fixes what must be measured, with what, and to what uncertainty (3.2).

3.1 Layer one — the characterization toolbox (survey)

No single instrument covers the capability vector; each trades resolution against environment, destructiveness, and the property it reports. The practical art is **correlative** — registering two or more techniques on the same specimen so that, e.g., atomic-resolution geometry and chemical identity are established together.

Technique	Reports	Resolution	Strengths	Limits / environment
STM	Surface topography + local density of states	atomic (<0.1 nm lateral)	Builds and verifies with the same probe; single-atom; electronic structure	Conductive surfaces; UHV/cryo; serial; surface-only
nc-AFM (CO-tip)	Force map; individual covalent bonds	sub-ångström	Direct bond imaging; works on insulators; force spectroscopy	UHV/cryo; serial; tip preparation sensitive
AC-(S)TEM / ptychography	Projected atomic columns; Z-contrast	<0.05 nm (sub-0.5 Å)	Sub-ångström even uncorrected via 4D-STEM ptychography; works on MOFs	Thin specimens; beam damage; projection (needs tomography for 3D)
Atom-probe tomography	3D position + chemical identity of single atoms	sub-nm (anisotropic)	Only technique giving 3D atom-by-atom composition	Destructive; needle specimens; reconstruction artifacts; partial detection
XRD / SAXS / GIWAXS	Lattice spacing, long-range order, distances	sub-ångström (d-spacing)	Statistical, non-destructive; ambient or in-situ; ensemble truth	Ensemble-averaged, not single-object; needs periodicity/contrast
XPS / EELS / EDS	Elemental + bonding state	EELS to atomic column	Chemical identity and oxidation state; pairs with (S)TEM	EELS needs TEM vacuum/thin sample; XPS surface-averaged
Cryo-EM / SPA	3D structure of soft assemblies	~2–3 Å routine	Solves DNA/protein scaffolds in near-native state	Soft matter only; ensemble averaging; not for stiff inorganic
DNA-PAINT / super-res.	Localization of labeled sites	~1–5 nm (sub-nm relative)	Counts and places individual addressable sites in DNA origami	Requires labels; optical, not atomic; soft scaffolds

Table 3. Characterization survey. Sub-ångström electron ptychography (to ~0.44 Å, even on radiation-sensitive MOFs) and CO-tip nc-AFM bond imaging are the two post-1995 leaps that make a material-agnostic precision standard enforceable.

3.2 Layer two — per-capability acceptance protocol

For an objective, trust-minimized judgment, each capability axis is tied to a primary metric, an accepted measurement method (with a permitted alternative for different platforms), a **metrology rule** capping measurement uncertainty relative to the claim, a sampling requirement, and an anti-spoofing control. The governing principle, borrowed from metrology practice, is that **the measurement uncertainty must be $\leq \frac{1}{3}$ of**

the tolerance being claimed — a claim of 0.1 nm placement is only admissible if the verifying instrument resolves to $\leq \sim 0.03$ nm on that specimen.

Capability	Primary metric	Method (primary / alt.)	Acceptance rule	Anti-spoofing control
Positional precision	RMS deviation from target coordinates over $\geq N$ features	AC-STEM ptychography or nc-AFM / APT for 3D	$u(\text{meas}) \leq \frac{1}{2}$ claimed tolerance; report full error budget	Judge-specified target revealed at test time; blind re-measure by 2nd lab
Atomic precision	Defects per placement (wrong site/species/missing)	Correlative STEM+EELS or APT over counted population	$\geq 10^4$ placements scored; CI stated; threshold pre-registered	Full population imaged, not a curated sub-field
Delivered force	Static force at site under load with selectivity held	AFM force spectroscopy; stiffness from resonance	Force measured during a real placement, not free-air; k cross-checked	Selectivity must persist under load; not feedback-reconstructed
Elemental generality	Species placed, confirmed; count of element classes	EELS/EDS or APT per element; XPS ensemble	Each claimed element independently confirmed on-site	Isotopic / fiducial labeling to exclude contamination
Product stiffness	Elastic modulus / phase identification	Nanoindentation or resonance; XRD phase ID	Modulus class stated with method; lattice confirmed	Tied to the actual built object, not a reference crystal
Throughput & duty	Reactions/s sustained over duration	In-situ time-resolved imaging / electrical readout	$\geq$ stated rate for $\geq$ stated seconds, logged continuously	Continuous record; no stitching of best intervals
Computing correctness	Exhaustive over all 65,536 input pairs	Direct read of output mechanism	100% correctness, all cases, single device	All inputs externally applied; output read without entrant interpretation
Computing energy	Energy per bit-operation	Calorimetric or electrical dissipation measurement	$\leq$ stated ceiling (e.g. 100 k_BT/bit-op); basis defined	Energy accounting independently auditable; define bit-op count per add

Table 4. Per-capability acceptance protocol. The unifying disciplines: (i) uncertainty $\leq \frac{1}{2}$ of the claim; (ii) judge-specified, test-time-revealed targets to force programmability over one-off artifacts; (iii) full-population statistics rather than curated sub-fields; (iv) independent second-lab replication for any apex claim.

Cross-cutting requirements

- Standardized reference targets.** A small library of judge-specified objects (a registry feature, a defined polyyne chain length, a dopant pattern) lets entrants on different platforms be scored on the same deliverable, and lets metrology be pre-validated against a known-answer specimen.
- Correlative registration.** Geometry, chemical identity, and force/stiffness must be established on the same specimen and spatially registered, not asserted from three different samples.
- Destructive-sample statistics.** Retain the original prize's strongest element — multiple copies surrendered for destructive analysis — but state explicitly whether the N copies are copies of the builder (repeatable capability) or of the built product (product quality); they certify different things.
- Pre-registration.** Thresholds, target specs, and the metrology protocol are fixed and published before the attempt, so a result cannot be reverse-fitted to whatever was achieved.

5. **Independent replication.** Any apex (atomic-precision) claim is confirmed by a second, unaffiliated laboratory using the published protocol.

4. Relationship to the Foresight prize

To be unambiguous: nothing here asks Foresight to change its prize. The Feynman Grand Prize keeps its role as the **stiff-covalent apex** — the hardest corner of Table 2, strong on every axis at once. What this document supplies is the surrounding *definition and measurement system* the field has lacked: a capability vocabulary in which scanning-probe, catalytic, scaffold, and crystalline approaches are all legible and comparable, and a characterization standard rigorous enough that claims made in any of them can be trusted. In that frame the original prize is recovered exactly — it is the simultaneous satisfaction of every axis, at diamondoid stiffness, verified to the Section 3.2 protocol.

The practical payoff is threefold. It lets a program (NDI's included) claim and demonstrate intermediate productive-nanosystem capability on named axes rather than waiting for an all-or-nothing apex; it gives funders and partners an objective scorecard to evaluate progress against; and it future-proofs the definition of molecular nanotechnology against the next approach, by defining success as a measurable capability rather than as any one machine.

Open items for the next revision

- **Force-reduction evidence (family B).** The ~100 pN-nN single-site-catalysis claim needs a measured or first-principles basis; it is currently the load-bearing assumption.
- **Energy ceiling justification.** Fix and defend a specific $k_{BT}/\text{bit-op}$ ceiling against a concrete molecular-logic device-physics estimate, and define “bit-operation” per 8-bit addition.
- **Reference-target library.** Specify the standardized judge-revealed objects and their pre-validated metrology.
- **Operating-environment classes.** Define admissible environments (UHV/cryo, ambient, aqueous) per capability, since they are not neutral across families.

References & notes

1. Foresight Institute, [Feynman Prizes](#) — Grand Prize (\$250,000), established 1996, unclaimed.
2. [Feynman Prize in Nanotechnology](#) (overview and Grand Prize criteria). Literal 1995 rules reproduced in Freitas & Merkle, *Kinematic Self-Replicating Machines* §4.9.4.
3. Underspecification / stiffness analysis: thermal-noise floor $k = k_{BT}/\langle x^2 \rangle \approx 0.41 \text{ N/m}$ at 0.1 nm RMS, 300 K; force-delivery $k \approx 60 \text{ N/m}$ for ~6 nN within 0.1 nm; ~145× gap. (NDI Definition Set, Doc 1, Appendix B; assumes 0.1 nm as RMS — a peak/3 σ reading raises the floor ~9×.)
4. Atomic-precision advanced manufacturing on Si: [Direct integration of APAM into middle-of-line silicon fabrication](#), Applied Physics Reviews 12, 041402 (2025); and [phosphine adsorption into engineered dangling bonds on H-Si \(STM + DFT\)](#), ACS Nano.
5. [Atomically precise mechanosynthesis of carbon structures on hydrogenated Si\(100\) by inverted-mode STM](#) (single-/multi-site C₂ donation; stepwise polyyne assembly via successive C–C bond formation).
6. [Atom-level enzyme active-site scaffolding using RFdiffusion2](#), Nature Methods (2025): 41/41 benchmark sites scaffolded vs 16/41 prior; active de novo enzymes from reaction mechanisms. Antibody design at atomic accuracy: [Atomically accurate de novo design of antibodies with RFdiffusion](#).
7. [Quantifying absolute addressability in DNA origami with molecular resolution](#) (sub-nm addressability via single-staple modification); [Position accuracy of gold nanoparticles on DNA origami by SAXS](#).
8. Bond imaging by nc-AFM with a CO-functionalized tip (Gross et al., pentacene, 2009) — field standard for direct covalent-bond visualization.
9. [Sub-0.5-Å-resolution ptychography in an uncorrected electron microscope](#), Science (2024): resolution to ~0.44 Å; [atomically resolved electron ptychography of radiation-sensitive MOFs](#), Nature Communications (2025). APT spatial-resolution limits: [Spatial resolution in atom probe tomography](#).

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